Learning a Safety-Aware CNN Cost Map for A*-Based Path Planning in Simulated Autonomous Driving Scenarios

Final Project Paper

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GitHub: https://github.com/xfang29/safety-aware-cnn-a-star-planning

Abstract

Classical A* search is interpretable and reliable, but its behavior depends on the traversal cost supplied to the search. A purely geometric objective favors the shortest collision-free route and does not directly encode driving preferences such as obstacle clearance, lane-centered motion, or road-boundary compliance. This project develops a hybrid learning-and-search framework in which a lightweight U-Net predicts a dense path-preference map from a simplified bird's-eye-view driving scene, while a conventional cost-aware A* planner remains responsible for producing the final connected and collision-free route. The system uses procedurally generated 64 x 64 scenes with five input channels, rule-based safety-aware A* demonstrations, Gaussian soft path labels, and disjoint training and validation splits. The U-Net contains 483,025 trainable parameters and was trained for 30 epochs with BCEWithLogitsLoss and Adam. A validation sweep selected a learned-cost weight of 2.0 before final testing. On 100 previously unseen final evaluation scenes, CNN-guided A* preserved a 100% planning success rate, reduced mean distance to the expert path from 1.708 to 0.361 grid cells, increased mean minimum obstacle clearance from 1.211 to 2.011 cells, and reduced mean lane deviation from 1.648 to 1.300 cells. The mean path length increased from 61.288 to 62.928 grid units, a 2.68% overhead. Paired Wilcoxon signed-rank tests strongly support improved expert similarity and obstacle clearance and also support reduced lane deviation. These results show that learned dense guidance can transfer expert safety preferences into A* while retaining hard feasibility constraints and incurring only a modest geometric-efficiency cost.

Keywords: A search; path planning; learned cost map; U-Net; autonomous driving; synthetic grid benchmark; safety-aware planning*

1. Introduction

Path planning is a central capability in autonomous robots and automated vehicles. A planner must connect a start state to a goal while respecting obstacles and environmental constraints. In a discrete grid representation, A* search is attractive because it is conceptually transparent, easy to verify, and optimal when its heuristic and cost assumptions are satisfied (Hart et al., 1968). However, the behavior of A* is determined by the objective it optimizes. When traversal cost includes only geometric distance and hard obstacle occupancy, the resulting route can pass close to obstacles, deviate unnecessarily from a lane center, or exhibit behavior that is graph-theoretically valid but undesirable for driving.

Autonomous-driving path quality is therefore not adequately described by collision avoidance alone. A useful route should maintain clearance from static obstacles, remain inside the drivable road region, prefer lane-centered motion when possible, and avoid excessive detours or abrupt local direction changes. These preferences can be encoded manually as weighted cost terms, but hand-tuned objectives become increasingly difficult to generalize as scene diversity grows. Conversely, a neural network can learn spatial preferences from demonstrations, but a network that directly outputs a path can produce disconnected or unsafe predictions and can be difficult to interpret.

This project adopts a hybrid strategy. A convolutional network predicts a dense spatial preference map, and a conventional A* planner converts that prediction into soft traversal costs while continuing to enforce hard road and obstacle constraints. The network therefore does not replace search; it provides a learned representation that guides an interpretable planner. The central research question is whether this learned guidance can reproduce safety-related expert behavior without sacrificing planning feasibility or causing large geometric detours.

The evaluation uses a controlled procedural benchmark rather than claiming immediate real-world driving readiness. Synthetic grid scenes permit reproducible variation in road curvature, obstacle placement, and scene difficulty through random seeds. Grid-based benchmarks are well established for pathfinding research (Sturtevant, 2012), while synthetic environments have also been used to study generalization of learned planners to unseen layouts (Tamar et al., 2016; Yonetani et al., 2021). The benchmark is used here to isolate the effect of learned spatial guidance under clearly defined constraints.

2. Related Work

2.1 Classical Grid Search and Benchmarks

Hart, Nilsson, and Raphael formalized A* as a heuristic graph-search method that combines the exact cost accumulated from the start with an estimate of the remaining cost to the goal (Hart et al., 1968). For an 8-connected grid with unit axial moves and square-root-of-two diagonal moves, octile distance provides a natural geometric heuristic. Standardized grid benchmarks subsequently enabled reproducible comparison of path length, runtime, and search effort across algorithms (Sturtevant, 2012). These works motivate the use of A* as the transparent search backbone in this study.

2.2 Learning Cost Maps and Planning Preferences

Learning a spatial cost function from demonstrations offers an alternative to manually selecting every planning penalty. Wulfmeier et al. (2017) learned large-scale navigation costs from expert driving demonstrations using deep inverse reinforcement learning, supporting the view that driving preferences can be represented as a dense cost map rather than direct low-level control. Perez-Higueras et al. (2018) similarly predicted a feasible path representation with a fully convolutional network and combined that prediction with a conventional planner. The broader principle is directly relevant here: a learned map can express behavioral preferences while a classical planner provides connectivity and collision checking.

2.3 Neural Planning and Learned Search Guidance

Value Iteration Networks embed approximate planning computation in a differentiable neural architecture and demonstrate generalization on path-planning domains (Tamar et al., 2016). Learning Heuristic Search via Imitation trains search guidance from expert decisions to reduce unnecessary search effort (Bhardwaj et al., 2017). Neural A* couples a convolutional encoder with a differentiable formulation of A* and learns a guidance map from expert paths (Yonetani et al., 2021). The present project is deliberately simpler: the search algorithm remains a conventional implementation and only the spatial preference map is learned. This separation reduces implementation risk and preserves a clear distinction between learned soft guidance and hard feasibility constraints.

2.4 U-Net for Dense Spatial Prediction

The network output is a dense 64 x 64 map rather than a class label. U-Net is suitable because its contracting path captures broader context while its expanding path and skip connections preserve spatial localization (Ronneberger et al., 2015). A lightweight U-Net can therefore map the five-channel scene tensor to a one-channel preference map without requiring a large model or real-world image corpus.

3. Research Project Problem

3.1 Problem Statement

Given a simplified bird's-eye-view grid containing a drivable-road mask, static obstacles, lane-center information, a start location, and a goal location, the objective is to generate a connected and collision-free path that balances geometric efficiency with driving-oriented safety preferences. Standard geometric A* serves as the primary baseline. The learned method predicts a dense map that lowers effective traversal cost in regions resembling expert behavior and raises the cost elsewhere. Obstacles and cells outside the road remain non-traversable regardless of network output.

3.2 Research Questions and Hypotheses

- H1: A CNN-predicted path-preference map will guide A* to produce paths that are closer to safety-aware expert paths than paths produced by standard geometric A*.
- H2: CNN-guided A* will improve safety-related metrics, particularly minimum obstacle clearance and lane-center adherence, relative to standard geometric A*.
- H3: CNN-guided A* will maintain comparable planning success with only a modest path-length overhead while improving path quality in cluttered driving-like scenes.

4. Method

Figure 1 summarizes the completed learning-and-search pipeline. Expert A* generates supervision, a lightweight U-Net learns a dense preference map, and the prediction is converted to a soft traversal-cost field used by a conventional A* search.

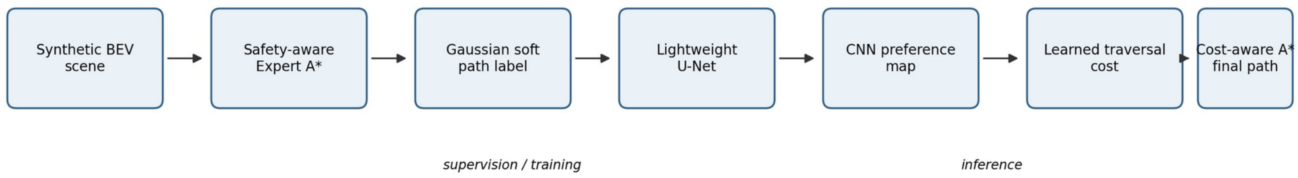

Figure 1. Completed hybrid learning-and-search pipeline.

4.1 Synthetic Driving-Scene Representation

Each sample is represented on a 64 x 64 grid. A mildly curved road centerline is generated procedurally, and the drivable road is formed by a fixed half-width around that centerline. Rectangular static obstacles are sampled inside the road. The generator reserves clear regions around the start and goal and rejects obstacle layouts for which no valid path exists. An 8-connected breadth-first search is used only as a feasibility check during scene generation, and diagonal corner cutting is prohibited.

Channel	Content	Interpretation
1	Road mask	Binary drivable region
2	Obstacle mask	Binary static-obstacle occupancy
3	Lane-center preference	Gaussian preference centered on the road centerline
4	Start map	Gaussian heatmap around the start cell
5	Goal map	Gaussian heatmap around the goal cell

Table 1. Five-channel model input representation.

4.2 Standard A* Baseline

The standard baseline searches free space using axial moves of cost 1 and diagonal moves of cost sqrt(2). Octile distance is used as the heuristic. The implementation returns the complete path, geometric length, expanded nodes, planning time, and success status. A separate path validator checks that the route begins at the specified start, reaches the goal, remains in free space, uses only adjacent grid moves, and never cuts diagonally through obstacle corners.

4.3 Safety-Aware Expert Planner

The expert planner uses the same hard feasibility constraints as standard A* but assigns additional traversal cost near obstacles and road boundaries and away from the lane center. For a free-space cell, the cost is

$$C(x,y) = 1 + w_o\, C_{obstacle}(x,y) + w_b\, C_{boundary}(x,y) + w_l\, C_{lane}(x,y).$$

The obstacle and boundary terms decay exponentially with Euclidean distance, while the lane term is one minus the lane-center preference map. The fixed expert weights are $w_o = 8$, $w_b = 4$, and $w_l = 2$. Obstacles and non-road cells receive infinite cost. For a move between adjacent cells u and v, edge cost is movement distance multiplied by the average of C(u) and C(v). Octile distance multiplied by the minimum finite cell cost is used as a lower-bound heuristic.

4.4 Gaussian Expert Labels

A one-pixel-wide expert path is highly imbalanced relative to the full grid and provides little learning signal to nearby cells. The binary path is therefore converted to a Gaussian soft target. If d(x,y) is the Euclidean distance from a cell to the nearest expert-path cell, the target is

$$P_{expert}(x,y) = \exp[-d(x,y)^2 / (2\sigma^2)], \quad \sigma = 2 \text{ grid cells.}$$

The label is zeroed on obstacles and outside the road. This creates a smooth corridor in which the exact expert path has value 1, nearby cells receive intermediate values, and distant cells approach zero.

4.5 Dataset Generation and Experimental Splits

The automated data pipeline generates a scene, constructs the rule-based expert cost map, runs and validates expert A*, builds the Gaussian label, and stores the five-channel input, soft label, binary path mask, start and goal coordinates, random seed, and expert path length. Training used 800 scenes generated from seeds 0-799. Validation used 100 disjoint scenes from seeds 10000-10099. An initial 100-scene model diagnostic split used seeds 20000-20099; after predictions from this split were inspected during development, it was treated as a development test set rather than as the final planning benchmark. A separate 100-scene final planner-evaluation set using seeds 30000-30099 was then generated and left untouched until the model architecture, training settings, and learned-cost weight were fixed.

4.6 Lightweight U-Net and Training

The model is a compact three-level U-Net. Each stage contains two 3 x 3 convolutions with batch normalization and ReLU activation. Three max-pooling operations reduce the spatial resolution from 64 x 64 to an 8 x 8 bottleneck; transposed convolutions and skip connections restore full resolution. The network returns raw logits, and sigmoid is applied only when a preference map is needed. The final configuration contains 483,025 trainable parameters.

Setting	Value
Input / target	5 x 64 x 64 / 1 x 64 x 64
Base channels	16 (encoder widths 16, 32, 64; bottleneck 128)
Trainable parameters	483,025
Loss	BCEWithLogitsLoss with Gaussian soft targets
Optimizer	Adam
Learning rate	1e-3
Batch size	32
Epochs	30
Random seed	4
Checkpoint rule	Lowest validation BCE; best epoch = 30
Best validation BCE	0.087787
Development test BCE	0.088473

Table 2. Final U-Net architecture and training configuration.

4.7 CNN-Guided Traversal Cost

At inference, the U-Net prediction P_CNN(x,y) in [0,1] is converted to a traversal penalty so that high predicted preference produces low cost and low preference produces high cost:

$$C\_CNN(x,y) = 1 + \text{lambda}\,[1 - P\_CNN(x,y)].$$

Cells outside free space are then overwritten with infinite cost, so the network cannot make an obstacle or non-road cell traversable. The scalar lambda controls how strongly the network affects A*. It was selected only on the validation set using candidate values 2, 4, 8, and 12.

5. Experimental Protocol

5.1 Baselines and Metrics

Final comparisons use three planners: standard geometric A*, the rule-based safety-aware expert A*, and CNN-guided A*. The primary evaluation metrics are success rate, geometric path length, minimum obstacle clearance, minimum road-boundary clearance, mean lane deviation, mean turn angle, expanded nodes, planning time, and symmetric mean nearest-neighbor distance to the expert path. The last measure is a symmetric Chamfer-style path distance: lower values mean stronger geometric agreement with the expert path.

5.2 Validation Selection of the Learned-Cost Weight

The learned-cost weight was chosen using the 100-scene validation set, not the final test set. Table 3 shows the validation sweep together with the standard and expert references. Lambda = 2 was selected because it captured most of the obstacle-clearance benefit while giving the shortest CNN-guided paths, the largest boundary clearance, the lowest lane deviation, and the lowest turn-angle cost among the tested learned weights.

Method	Success	Length	Obs. clear	Boundary	Lane dev.	Turn
Standard	1.000	60.934	1.327	7.048	1.579	0.061
Expert	1.000	62.749	2.496	7.036	1.119	0.139
CNN lambda=2	1.000	62.315	2.122	7.304	1.030	0.155
CNN lambda=4	1.000	62.485	2.181	7.240	1.047	0.165
CNN lambda=8	1.000	62.621	2.208	7.201	1.054	0.172
CNN lambda=12	1.000	62.679	2.218	7.197	1.054	0.176

Table 3. Validation-set learned-cost weight sweep. Distances are in grid cells; turn angle is in radians.

5.3 Statistical Analysis

All final planners are evaluated on the same 100 held-out scenes, yielding paired observations. H1 uses expert-path distance; H2 uses minimum obstacle clearance and mean lane deviation; H3 is evaluated descriptively through success rate and path-length overhead. Because path metrics are discrete, skewed, and frequently tied, one-sided paired Wilcoxon signed-rank tests are used for the directional H1 and H2 hypotheses. A two-sided Wilcoxon test is used to characterize path-length change. For reporting effect magnitude, 95% bootstrap confidence intervals for the mean paired differences were computed from the final per-scene records using 20,000 resamples and a fixed random seed of 4. Statistical significance is interpreted together with effect size rather than as a substitute for practical relevance.

6. Results

6.1 U-Net Training and Dense Prediction Quality

Training and validation losses decreased smoothly throughout all 30 epochs and remained closely matched, with the best validation BCE of 0.087787 at epoch 30. The development test BCE was 0.088473, indicating similar model-level error on unseen generated scenes. Figure 2 shows that the learning curves do not exhibit a widening train-validation gap over the selected training horizon.

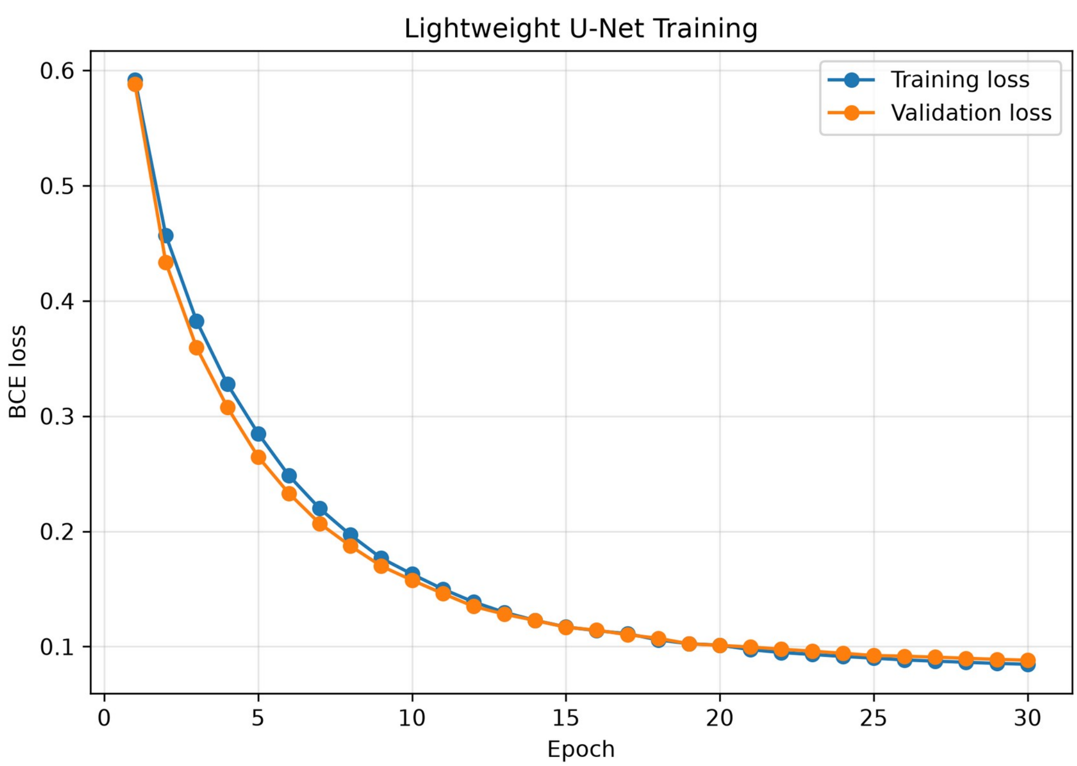

Figure 2. Training and validation BCE loss for the lightweight U-Net over 30 epochs.

Representative prediction maps in Figure 3 show that the network follows road curvature and changes its high-preference corridor in response to scene geometry and obstacle placement. The predictions are smoother and somewhat wider than the one-cell expert trajectories, which is consistent with the Gaussian target construction and is desirable for use as soft search guidance rather than direct path output.

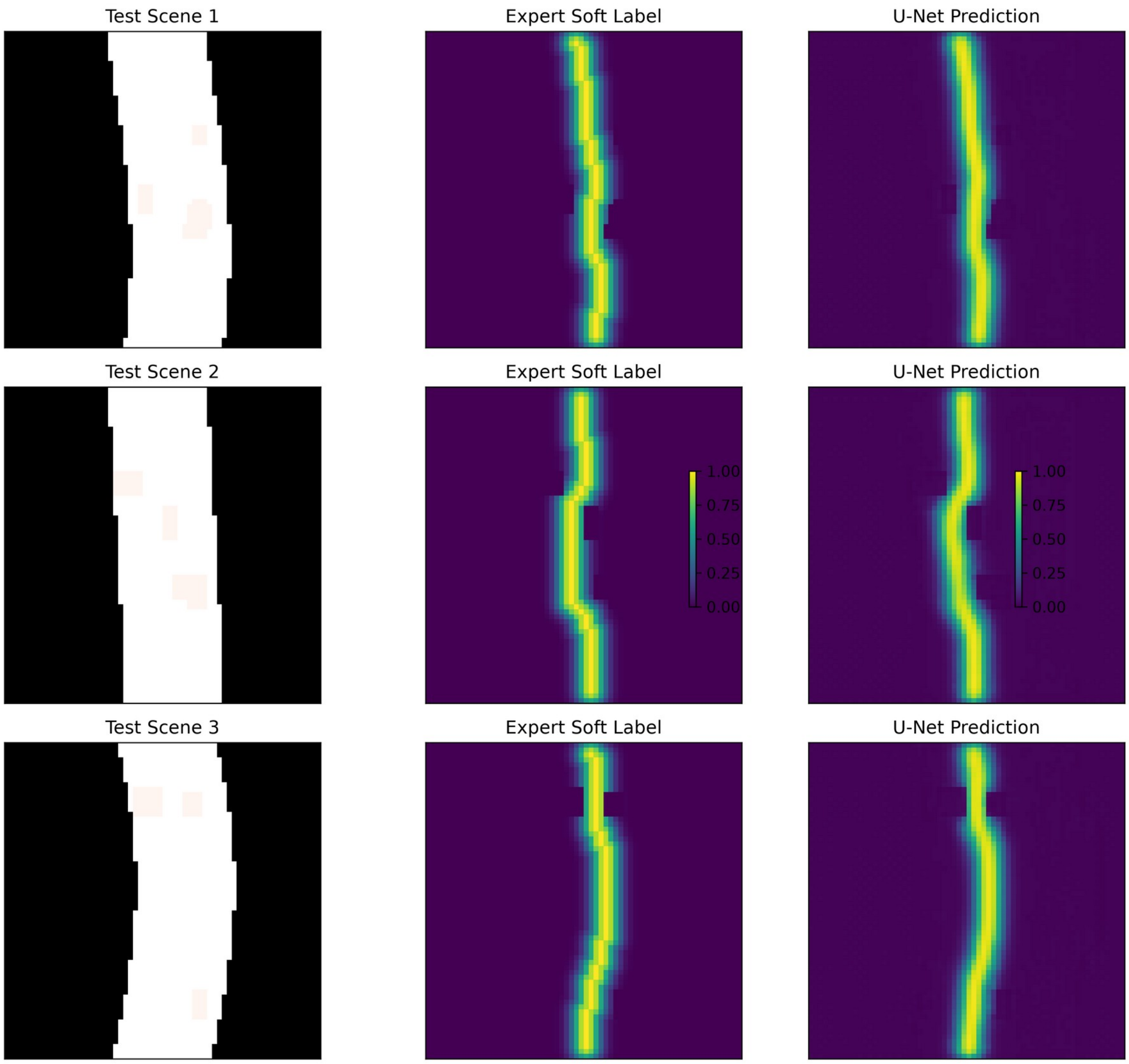

Figure 3. Representative development-test scenes, Gaussian expert supervision maps, and U-Net predictions.

6.2 Final Held-Out Planner Performance

Table 4 reports the final 100-scene evaluation on seeds 30000-30099. All three planners achieved a 100% success rate. Relative to standard A*, CNN-guided A* reduced the average distance to the expert path by 78.9%, increased minimum obstacle clearance by 66.1%, and reduced mean lane deviation by 21.1%. Mean path length increased by 2.68%. Boundary clearance changed by less than 1%, while mean turn angle, expanded nodes, and runtime increased because the learned cost reshapes the search objective and introduces additional local directional changes. Even with this overhead, mean CNN-guided planning time remained approximately 1.62 ms on the evaluation machine.

Metric	Standard A*	Expert A*	CNN-guided A*	CNN vs. Std.
Success rate	1.000	1.000	1.000	No change
Path length	61.288	63.257	62.928	+2.68%
Min. obstacle clearance	1.211	2.280	2.011	+66.1%
Min. boundary clearance	6.792	6.585	6.726	-1.0%
Mean lane deviation	1.648	1.338	1.300	-21.1%
Mean turn angle	0.068	0.144	0.157	+130.4%
Expanded nodes	179.07	896.95	298.88	+66.9%
Planning time (ms)	0.705	4.361	1.622	+130.0%

Metric	Standard A*	Expert A*	CNN-guided A*	CNN vs. Std.
Distance to expert	1.708	0.000	0.361	-78.9%

Table 4. Final held-out comparison over 100 scenes. Distance metrics are measured in grid cells; turn angle is in radians.

6.3 Representative Path Behavior

Figure 4 illustrates the qualitative behavior underlying the aggregate metrics. Standard A* follows the geometric shortcut, the expert follows the hand-designed safety preferences more strongly, and CNN-guided A* shifts toward the expert corridor while retaining a path length close to the geometric baseline. This example is representative of the intended hybrid behavior: the network changes where A* prefers to search but does not replace hard connectivity and collision constraints.

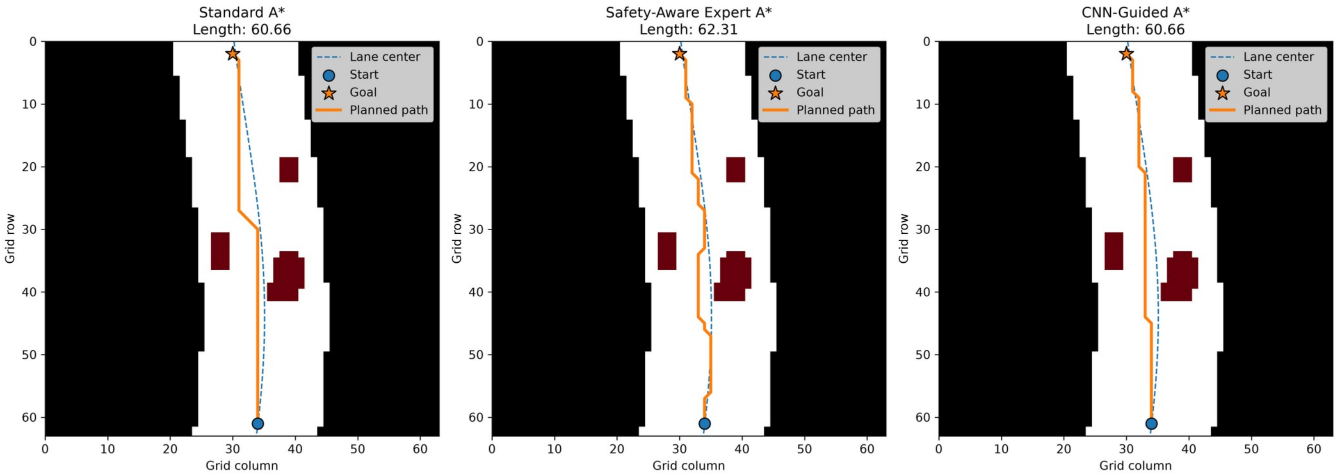

Figure 4. Representative held-out comparison of standard A, safety-aware expert A*, and CNN-guided A*.*

6.4 Hypothesis Tests

Table 5 summarizes the paired statistical tests. H1 is strongly supported: CNN-guided paths are substantially closer to the expert than standard A* paths. H2 is also supported by both primary safety proxies: obstacle clearance increases and lane deviation decreases. H3 is supported in the practical engineering sense rather than as a claim of statistical equivalence. The CNN-guided planner preserved 100% success, but its path length was systematically longer; the average increase was only 2.68%, so the efficiency penalty is modest relative to the safety-related gains.

Hypothesis / metric	Mean paired diff.	95% bootstrap CI	Wilcoxon W	p-value	Decision
H1: distance to expert	-1.347	[-1.600, -1.120]	20.0	3.56e-18	Supported
H2a: obstacle clearance	+0.800	[+0.662, +0.940]	2252.5	2.74e-13	Supported
H2b: lane deviation	-0.347	[-0.533, -0.164]	1510.0	2.42e-4	Supported
H3: path length	+1.640	[+1.383, +1.901]	5.0	1.31e-16	Modest trade-off

Table 5. Paired final-test statistics for CNN-guided A relative to standard A*. Directional tests are one-sided for H1/H2 and two-sided for path length.*

7. Discussion

The strongest result is that the learned preference map transfers expert structure into a conventional planner. The final distance-to-expert metric decreases from 1.708 to 0.361 cells, an approximately 78.9% reduction. This is important because the network is not evaluated only by image-space loss: its learned output materially changes the route produced by A*. The result supports the central design choice of predicting a dense guidance field rather than a complete path.

Safety-related behavior also improves in a consistent direction. Minimum obstacle clearance rises from 1.211 to 2.011 cells, and mean lane deviation falls from 1.648 to 1.300 cells. Interestingly, CNN-guided A* achieves slightly lower mean lane deviation than the expert in the final set, while retaining lower obstacle clearance than the expert. This does not imply that the network is universally better than its teacher. Instead, the Gaussian target and convolutional prediction smooth the expert demonstrations, which can shift the final compromise among lane, obstacle, and geometric costs.

The learned method does not dominate standard A* on every metric. Mean path length is 2.68% larger, turn angle more than doubles, and search time increases from about 0.71 ms to 1.62 ms. The path-length difference is statistically significant, so the paper does not claim equivalence. The engineering claim is narrower: the observed safety-proxy improvements are obtained without any loss of planning success and with a small geometric-length penalty. The runtime remains small in this synthetic 64 x 64 setting.

The expert remains useful as a reference rather than as an absolute optimum. It achieves the largest obstacle clearance but expands far more nodes and has the longest paths among the learned and geometric methods. CNN guidance captures much of the expert behavior at substantially lower search effort than the rule-based expert. This illustrates one possible benefit of learning a dense spatial prior: expert preferences can be amortized into a fast prediction and then consumed by a standard planner.

8. Limitations and Threats to Validity

- The benchmark is synthetic and two-dimensional. It does not model vehicle dynamics, nonholonomic constraints, dynamic obstacles, perception uncertainty, traffic rules, or interaction with other agents.
- The operational definition of safety is inherited from a hand-designed expert cost map. The network learns this definition rather than a universal or human-validated driving policy.
- The road generator covers a limited family of mildly curved roads and rectangular obstacles. Generalization claims are therefore restricted to the tested procedural distribution.
- Distances are measured in grid cells rather than physical meters, and the discrete turn-angle metric is not equivalent to vehicle curvature or steering comfort.
- The final test contains 100 scenes. This is sufficient for a course-scale paired comparison but remains small relative to production learning systems and does not establish real-world robustness.
- No outright planning failures occurred in the final test because hard feasibility constraints are retained and the scene generator rejects infeasible layouts. Consequently, failure analysis focuses on behavioral trade-offs such as higher turn angle and search effort rather than collision failures.
- The model-level development test split was inspected during training diagnostics. To avoid contaminating the final planning evaluation, a new seed range was generated and reserved for the final planner comparison.

9. Conclusion

This project demonstrates a complete hybrid learning-and-search pipeline for safety-aware A* planning in simulated driving-like grids. A rule-based expert produces Gaussian soft supervision, a lightweight U-Net learns a dense path-preference map, and the prediction is converted into a soft traversal cost while hard obstacle and road constraints remain enforced by A*. On 100 fully held-out final scenes, CNN-guided A* maintains 100% success, moves substantially closer to expert trajectories, increases obstacle clearance, and reduces lane deviation. These improvements come with a statistically significant but modest 2.68% increase in path length and higher local turn and search-effort metrics. The findings therefore support the core hypothesis that learned spatial guidance can improve the qualitative behavior of an interpretable planner without replacing the planner itself. Future work should extend the benchmark to richer road geometries, dynamic obstacles, orientation-aware or nonholonomic planning, learned multi-objective costs, and higher-fidelity simulation before any claim of real-world autonomous-driving applicability.

Reproducibility Note

The project implementation is available at https://github.com/xfang29/safety-aware-cnn-a-star-planning. Reusable Python modules are organized in src/, executable notebooks are in notebooks/, the trained checkpoint is in models/best_unet.pt, and final per-scene results, summary tables, statistical outputs, and figures are in results/. Generated training arrays are intentionally excluded from the repository because notebooks/03_dataset_generation.ipynb reproduces them from deterministic seed ranges. The final evaluation records are retained in CSV form so that reported summary statistics and additional paired analyses can be reproduced without rerunning the planners.

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